

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

*I **Bharath P(192511282)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Bharath

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

- (a)** Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b)** Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c)** If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CASE STUDY 1 — Smart Medical Diagnosis System

The knowledge base contains three diagnostic rules and three facts about Patient A.

(a) Representation using Propositional Logic

Define the symbols

Let:

- **F** = Patient A has Fever
- **C** = Patient A has Cough
- **B** = Patient A has Breathlessness
- **Flu** = Patient A has Flu
- **M** = Patient A has Measles
- **P** = Patient A has Pneumonia

Rules

Rule 1: If Fever AND Cough, then Flu.

Rule 2: If Fever AND Rash, then Measles.

Let **R** = Patient A has Rash.

Rule 3: If Cough AND Breathlessness, then Pneumonia.

Facts

Therefore, the complete knowledge base is:

(b) Forward Chaining

Forward chaining starts with the available facts and repeatedly applies rules whose premises are satisfied.

Initial facts

Step 1 — Apply Rule 1

Rule:

Since:

both conditions are satisfied.

Therefore:

Step 2 — Apply Rule 2

Rule:

We know:

but there is **no fact that Rash = True**.

Therefore, Rule 2 cannot fire.

Step 3 — Apply Rule 3

Rule:

We know:

and

Therefore:

Final conclusions

Forward-chaining sequence

Hence:

(c) Backward Chaining for Pneumonia

Goal

Find a rule whose conclusion is Pneumonia.

Rule 3:

Therefore, the goal is reduced into two sub-goals:

↓

Now check the facts:

Both sub-goals are satisfied.

Therefore:

Goal reduction tree

Goal: Pneumonia(A)

|

Rule 3: $C \wedge B \rightarrow P$

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Cough(A) $\wedge$

Breathlessness(A)

/

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✓

✓

Cough = True Breathlessness = True

Hence, the goal is successfully proved.

CASE STUDY 2 — Autonomous Traffic Management Agent

The second case uses **First-Order Logic (FOL)**, unification and resolution.

(a) Unification of Rule 1

Rule 1

Facts:

We need to match:

with:

Therefore:

Substitution

After substitution, Rule 1 becomes:

Both premises are facts.

Therefore:

MGU

This is the **Most General Unifier (MGU)**.

(b) Resolution proof that Proceed(CarA) is true

Relevant rules:

Rule 1

CNF:

Rule 2

and:

CNF:

Rule 3

CNF:

Facts

Step 1: Derive ClearPath(Ambulance)

From:

and:

and:

we obtain:

Step 2: Derive GreenSignal(Ambulance)

Rule 2:

Substitute:

giving:

Since:

we derive:

Step 3: Derive Proceed(CarA)

Rule 3:

Substitute:

Therefore:

We have:

Therefore:

Resolution chain

Vehicle(Ambulance)

+

EmergencyType(Ambulance)

↓

ClearPath(Ambulance)

↓

GreenSignal(Ambulance)

Vehicle(CarA)

+

Behind(CarA,Ambulance)

+

GreenSignal(Ambulance)



Proceed(CarA)

Hence:

(c) Two simultaneous emergency vehicles

The current KB can handle **multiple vehicles individually**, because Rule 1 is universally quantified:

However, it does not explicitly represent interactions between **two simultaneous emergency vehicles**.

For example, suppose we have:

Rule 1 can independently derive:

and:

But additional information is needed to determine:

- Which ambulance has priority.
- Whether both can use the same road.
- Whether their paths conflict.
- Which vehicle should receive the green signal first.
- Whether one vehicle is behind another emergency vehicle.

Additional rules needed

For example:

and:

Also, facts such as:

and:

may be required.

Conclusion

It needs **priority, conflict, route and coordination rules**.

CASE STUDY 3 — Agricultural Expert Reasoning System

The third case asks for CNF conversion and resolution refutation.

Let:

- **S** = SoilDry
- **I** = IrrigationNeeded
- **W** = CropWheat
- **D** = ApplyDripMethod

- **R** = CropAtRisk

(a) Convert to CNF

P1

Given:

Symbolically:

Step 1 — Eliminate implication

Therefore:

This is already in CNF.

P2

Given:

Eliminate implication:

Using De Morgan's law:

Therefore:

P3

Given:

Eliminate implication:

Using De Morgan's law:

Therefore:

Facts

becomes:

and:

becomes:

Complete CNF knowledge base

(b) Resolution Refutation to prove ApplyDripMethod

Goal

Negate the goal:

Add this to the KB.

Therefore:

Now perform resolution.

Resolution 1

Take:

and:

Resolving and :

Resolution 2

Take:

and:

Resolving and :

So:

Resolution 3

Take:

and:

Resolving and :

Resolution 4

We have:

and the negated goal:

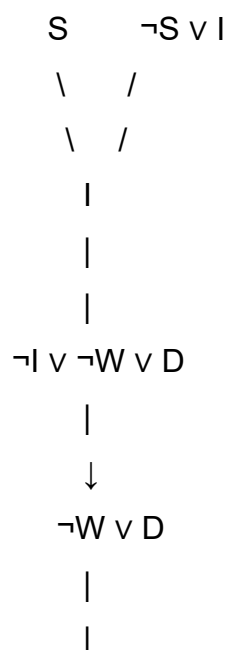
Resolving:

gives:

The empty clause represents a contradiction.

Therefore the negated goal is impossible, and hence:

Resolution proof tree



W
 $\downarrow$
 D
 $|$
 $|$
 $\neg D$
 $\downarrow$
 $\square$

Therefore:

(c) Remove SoilDry

Now remove:

from the knowledge base.

Therefore, the fact:

is no longer available.

The remaining facts are:

and the rules:

Step 1

Previously we had:

and:

which allowed us to derive:

But now is absent.

Therefore:

Step 2

Without , Rule P2 cannot establish:

Therefore:

Step 3 — CropAtRisk

P3 is:

To derive , we need:

and:

We have:

but the KB does **not explicitly provide**:

Therefore:

Final conclusion

Removing SoilDry breaks the inference chain:

Without :

CASE STUDY 4 — Wumpus World Knowledge Agent

The fourth case gives three percepts: **Stench at [1,2], Breeze at [1,1], and Glitter at [2,2]**.

Let:

- = Stench at
- = Breeze at
- = Wumpus at
- = Pit at
- = Gold at
- = Cell is safe

(a) Belief state and Modus Ponens

Initial percepts

The agent observes:

Therefore the initial belief state is:

Rule 1 — Stench

Given:

Since:

by Modus Ponens:

The adjacent cells are:

Therefore possible Wumpus locations are:

Rule 2 — Breeze

Given:

Since:

we derive:

Adjacent cells include:

Therefore possible Pit locations are:

Rule 3 — Glitter

Given:

Since:

we derive:

Belief state

Thus:

(b) Forward Chaining — Possible Wumpus and Pit Locations

Wumpus

Initial fact:

Rule:

Therefore:

Possible neighboring cells:

Hence:

Pit

Initial fact:

Rule:

Therefore:

Possible neighboring cells:

Hence:

Gold

Initial fact:

Rule:

Therefore:

(c) Is the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ safe?

The proposed path is:

We must evaluate every cell.

Cell $[1,1]$

There is a breeze at:

which indicates that a pit may be adjacent.

However, the current knowledge does not establish that $[1,1]$ itself contains a pit.

So:

Cell $[1,2]$

We have:

This means a Wumpus is adjacent to [1,2].

Also, because:

the possible pit locations include:

Therefore [1,2] has a possible pit, and the Wumpus information also makes the surrounding region dangerous.

Thus:

Cell [2,2]

We know:

However, because of the stench at [1,2], [2,2] is one of the possible Wumpus locations:

Therefore:

Final conclusion

The path:

is **not guaranteed to be safe** based on the available knowledge.

The reason is that the current KB leaves possible hazards at cells along the route, particularly **[1,2] and [2,2]**.
